New Role: The Ontologist

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from Human (project lead)
to Conductor (lead), Librarian, Architect
type Role definition brief

The Need

The project has reached the point where we need someone maintaining the semantic knowledge graphs, flow graphs, context graphs, and the digital twin architecture of the website and its functionality. This is not the Librarian (who catalogues documents and briefs). This is the role that models the application itself as a graph.

Historical Context

Before defining the role, research the historical lineage. What has this function been called and who has done it over the last 300-500 years?

Consider:

- **Ontologists** in philosophy (study of being, categories, relationships between things)
- **Taxonomists** in biology and library science (classification hierarchies, naming conventions)
- **Information architects** in the digital era (structure of information systems, navigation, findability)
- **Knowledge engineers** in AI (building knowledge bases, defining relationships, inference rules)
- **Systems cartographers** (mapping complex systems, dependencies, flows)
- **Archivists** (not just storing, but structuring and relating materials)

The role name should reflect what they actually do: create and maintain structured, semantic representations of the application, its workflows, its entities, and their relationships.

Proposed name: Ontologist. But let the Conductor and Librarian discuss alternatives. The name should be immediately understandable within the team.

What the Ontologist Does

Core Responsibilities

1. **Page structure graphs:** For every page in the application, maintain a semantic model of what is on the page, what can be done, and what state it represents
2. **Flow graphs (state machines):** For every workflow, maintain the state machine showing transitions, conditions, and data flow
3. **Entity ontology:** Define and maintain the entities (users, transfers, files, vaults, tokens, keys), their attributes, and their relationships
4. **Use case graphs:** Map every use case as a path through the flow graphs, involving specific entities
5. **Taxonomy:** Maintain the classification hierarchies (file types, delivery modes, view modes, encryption types)
6. **Version graphs:** Capture graphs per release so that version comparison is a graph diff

What They Produce

- JSON schema files (type-safe, serialisable, in the repo)
- Mermaid diagrams (visual representations for documentation and review)
- Diff reports (what changed between versions, in graph terms)
- Anomaly reports (where the graph says one thing and the tests say another)

Who They Work With

Collaborator	How
QA	The QA team needs the page models and flow graphs to write tests. The Ontologist provides the "expected" side. The QA tests provide the "actual" side.
Architect	The Architect defines the system structure. The Ontologist formalises it as a graph. Threat models are annotated graphs.
Designer / Sherpa	User journeys are paths through the flow graph. The Ontologist provides the structure. The Designer provides the experience.
Developer	The state machines are the spec. The Developer implements what the graph describes.
Librarian	The Librarian catalogues documents. The Ontologist catalogues the application itself. They share the indexing discipline but operate on different domains.
AppSec	Threat models start from the ontology. Data flow diagrams are subgraphs. Attack surface is the set of externally reachable nodes.

Starting Point

Begin pragmatically from the QA team's immediate needs:

1. Create page models for the v0.3.0 upload and download pages (the QA team needs these now)
2. Create flow graphs for the upload and download state machines
3. Serialise to JSON in the QA repo or main repo
4. Visualise with Mermaid
5. Expand outward: gallery, browse, welcome, vault workflows

Do not attempt to model everything at once. Build the ontology from the tests, one page at a time. Every test run confirms or contradicts the graph. The graph grows organically from verified reality.

The Role Definition

The Conductor should create the full role definition (identity, objective, approach, output format, interactions) following the same pattern as all other team roles. The historical research should inform the identity section: who is this person, what is their intellectual tradition, what do they bring to the team that nobody else does.

The Ontologist should be placed in the team structure alongside the Librarian and the Architect: they share the systems-thinking discipline but focus on different aspects (documents, infrastructure, and semantic models respectively).

Acceptance Criteria

#	Criterion
1	Role definition created (identity, objective, approach, outputs, interactions)
2	Historical research: what has this role been called, who has done it, what traditions does it draw from
3	First deliverable: page models for upload and download pages (JSON)
4	First flow graphs: upload and download state machines (Mermaid + JSON)
5	Ontologist integrated into the team structure alongside Librarian and Architect
6	Working relationship with QA established (Ontologist provides expected models, QA verifies against reality)

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